Online Training – Advanced session
February 2025

Turbulence modeling in OpenFOAM: Guided Tutorials

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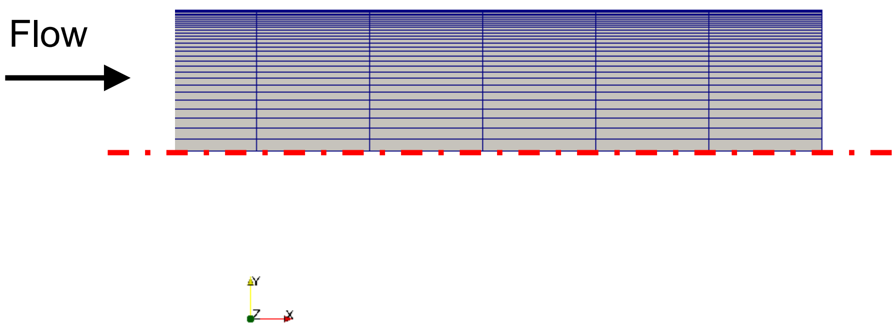
Guided tutorials

- **Guided tutorial 1.** Laminar-Turbulent pipe.
- This case is ready to run.
- The case is located in the directory:

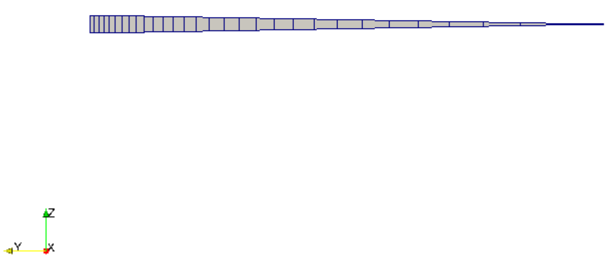
```
$TM/turbulence/GT1/
```

- In the case directory, you will find a few scripts with the extension .sh, namely, `run_all.sh`, `run_mesh.sh`, `run_sampling.sh`, `run_solver.sh`, and so on.
- These scripts can be used to run the case automatically by typing in the terminal, for example,
  - `$> sh run_solver`
- These scripts are human-readable, and we highly recommend you open them, get familiar with the steps, and type the commands in the terminal. In this way, you will get used with the command line interface and OpenFOAM commands.
- If you are already comfortable with OpenFOAM, run the cases automatically using these scripts.
- In the case directory, you will also find the `README.FIRST` file. In this file, you will find some additional comments.

Guided tutorial 1 – Laminar-Turbulent pipe

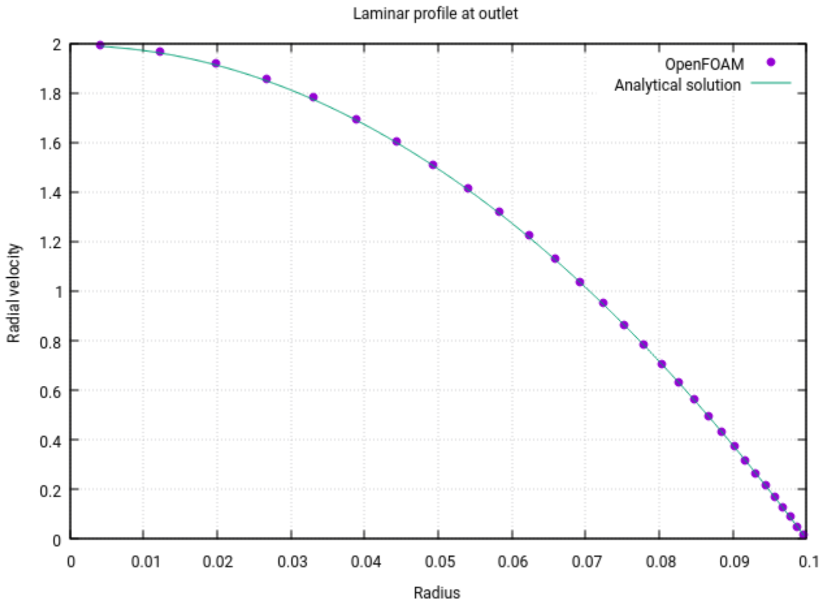


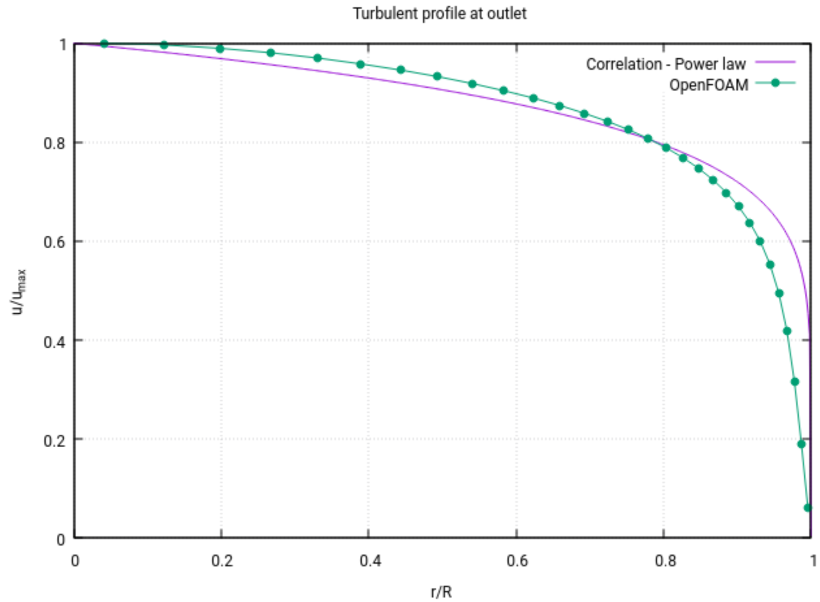
Circular wedge



- In this guided tutorial we will simulate a laminar and a turbulent flow in a circular pipe.
- We will use axial symmetry and a pure 2D approach.
- This problem has plenty of experimental data and analytical correlations for validation.

Guided tutorial 1 – Laminar-Turbulent pipe





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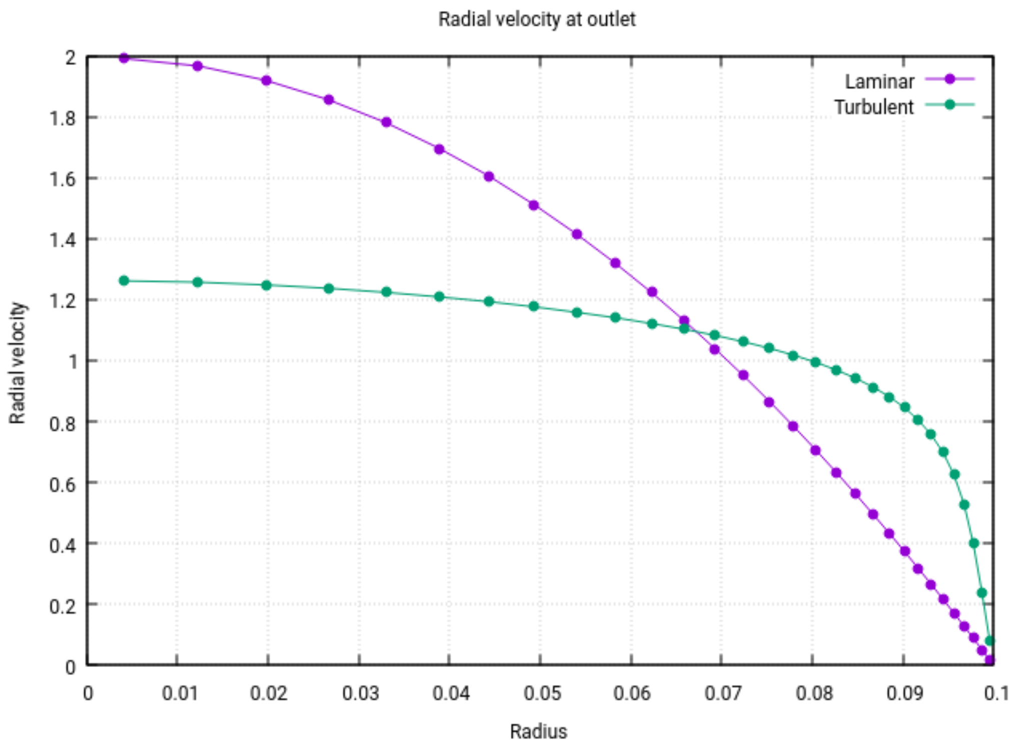

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- To set a turbulent model in OpenFOAM, you will need to modify the dictionary *momentumTransport*. This dictionary is located in the directory **constant**.

```
simulationType   RAS;          <- RANS type simulation

RAS                            <- RANS sub-dictionary
{
    RASModel        kOmegaSST;     <- RANS model to use

    turbulence      on;            <- Turn on/off turbulence.  Runtime modifiable

    printCoeffs     on;            <- Print model coefficients
}
```

- In this dictionary you choose what kind of approach you want to use (keyword **simulationType**): laminar, RAS, or LES.
- In you choose a RAS or LES approach, you will need to define a sub-dictionary with the options specific to the selected model.

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- To option **printCoeffs** in the dictionary *momentumTransport* will print all the constants/coefficients used in the model (you can also read the source code to get this information).
- In this case, you will get the following information in the output screen,

```
RASModel          kOmegaSST;
turbulence        on;
printCoeffs       on;
alphaK2           1;
alphaK1           0.85;
alphaOmega1       0.5;
alphaOmega2       0.856;
gamma1            0.555555555556;
gamma2            0.44;
beta1             0.075;
beta2             0.0828;
betaStar          0.09;
a1                0.31;
b1                1;
c1                10;
F3                false;
```

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- So, if you want to change one or more of these constants/coefficients just add the new entry in the dictionary *momentumTransport*, in the sub-dictionary RAS (after the keyword **printCoeffs**), as follows,

```
simulationType  RAS;

RAS
{
    RASModel        kOmegaSST;
    turbulence      on;
    printCoeffs     on;

    alphaK1         8.31445;
    alphaK2         3.14159;

}
```

New model coefficients

- This works for all RANS and LES models.
- No need to say that you need to know what are you doing.

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- When dealing with turbulence, setting boundary and initial conditions is of paramount importance.
- Our best advice is to get familiar with the theory.
- In the NASA Turbulence Modeling Resource site, you will find plenty of information:

https://turbmodels.larc.nasa.gov/

- If you are feeling lazy and you do not have good estimates, you can use our calculator for the estimation of turbulence properties (use with caution):

http://www.wolfdynamics.com/tools.html?id=110

- If you want to estimate the y+ value or get an idea of where to put the first mesh vertex or cell center normal to the wall, you can use our calculator:

http://www.wolfdynamics.com/tools.html?id=2

- Remember to choose the near the wall treatment.

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- The initial value for the turbulent kinetic energy k can be found as follows

$$k_{farfield} = \frac{3}{2}(UI)^2$$

- The initial value for the specific kinetic energy ω can be found as follows

$$\omega_{farfield} = \frac{\rho k}{\mu}\left(\frac{\mu_t}{\mu}\right)^{-1}$$

- If you need to, the initial value for the turbulent dissipation ϵ can be found as follows

$$\epsilon_{farfield} = \frac{C_\mu k^2}{\nu \cdot (\mu_t/\mu)}$$ where $$C_\mu = 0.09$$

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- If you are totally lost, you can use the following reference values.
- They work most of the times, but it is a good idea to have some experimental data or a better initial estimate.

	Low	Medium	High
I	1.0 %	5.0 %	10.0 %
μ_t/μ	1	10	100

- Where μ_t/μ is the viscosity ratio and $I = u'/\bar{u}$ is the turbulence intensity.
- By the way, use these guidelines for external aerodynamics only.
- Use these values only if you do not have better estimates.

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- When it comes to near-wall treatment, you have three options: use wall functions ($30 < y^+ < 300$), use y^+ insensitive wall functions ($1 < y^+ < 300$), and resolve the boundary layer ($y^+ < 5\text{-}6$).
- Hereafter you will find a list of typical wall functions boundary conditions:

Field	Wall functions – High RE	Resolved BL – Low RE
nut	**nut(–)WallFunction*** or **nutUSpaldingWallFunction**** (with 0 or a small number)	**nutUSpaldingWallFunction****, **nutkWallFunction**, **nutUWallFunction**, **nutLowReWallFunction** or **fixedValue***** (with 0 or a small number)
k, q, R	**kqRWallFunction**** (with inlet value or a small number) $k_{wall} = k$	**kqRWallFunction**** or **kLowReWallFunction** (with inlet value, 0, or a small number) or **fixedValue***** (with 0 or a small number)
epsilon	**epsilonWallFunction** (with inlet value) $\epsilon_{wall} = \epsilon$	**epsilonWallFunction** (with 0 or a small number) or **zeroGradient***** or **fixedValue***** (with 0 or a small number)
omega	**omegaWallFunction**** (with a large number) $\omega_{wall} = 10\frac{6\nu}{\beta y^2} \qquad \beta = 0.075$	**omegaWallFunction**** or **fixedValue***** (both with a large number)
nuTilda	–	**fixedValue** (one to ten times the molecular viscosity, a small number, or 0)

* nutUWallFunction or nutkWallfunction
** Recommended options for y^+ insensitive treatment (continuous wall functions)
*** Will disable wall functions. The equations will be integrated down to the viscous sublayer with no damping or corrections.

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- Typical **nut** wall functions boundary conditions are:
    - **nutkRoughWallFunction** (high-Re)
    - **nutkWallFunction** (high-Re and low-Re - y^+ insensitive)
    - **nutLowReWallFunction** (low-Re)
    - **nutUWallFunction** (high-Re and low-Re - y^+ insensitive)
    - **nutUSpaldingWallFunction** (high-Re and low-Re - y^+ insensitive using Spalding continuous function)
- A low-Re turbulence model solves the boundary layer ($y^+ < 5\text{-}6$).
- A high-Re turbulence model uses wall functions ($30 < y^+ < 300$).
- Low-Re means that the local Re is low (as in the boundary layer), and high-Re means the local Re is high. These definitions are not related to the global Re.
- Remember, in order to use wall functions, you need to define the patch type as a **wall** in the dictionary `constant/polyMesh/boundary`.
- Also, if you are dealing with moving walls, do not forget to use boundary conditions that take into account the grid/patch velocity.

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- The boundary and initial conditions are set in the dictionary files located in the `0` folder.
- As we are using the $k-\omega$ model, we will need the following dictionary files: $\mathtt{k}$, $\mathtt{omega}$, and $\mathtt{nut}$.
- k and ω are the additional closure equations of the turbulence model, and nut is the turbulent viscosity.
- The initial conditions can be calculated using the recommended formulas, but if you have better estimates feel free to use them.
- In this case we will use the following boundary conditions at the walls: for k we use the boundary condition **kqRWallFunction**, for ω we use **omegaWallFunction**, and for nut we use **nutUSpaldingWallFunction**.
- The **calculated** boundary condition used in the $\mathtt{nut}$ dictionary, means that its value is computed from k and ω.
- This case setup will use y^+ insensitive wall functions.
- This is a pretty standard setup for the $k-\omega$ model.

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- Reminder:

  - The flow is incompressible.
  - The pipe radius and length are 0.1 m and 8.0 m, respectively.
  - And we are targeting for a Re = 100 (laminar flow), and Re = 10000 (turbulent flow).

$$Re = \frac{\rho \times U \times D}{\mu} = \frac{U \times D}{\nu}$$

$$\nu = \frac{\mu}{\rho}$$

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- Reminder:

  - To compute the wall distance units y+, you can use the following equation

$$y^+ = \frac{\rho \times U_\tau \times y}{\mu} = \frac{U_\tau \times y}{\nu}$$

  - Where *y* is the distance to the first cell center normal to the wall, and U_τ is the friction velocity and is equal to

$$U_\tau = \sqrt{\frac{\tau_w}{\rho}}$$

where τ_w is the wall shear stresses.

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- We are going to use the following modular solver: `incompressibleFluid`.
- This case is rather simple, but we will use it to explain many features used in OpenFOAM when dealing with turbulence.
- We will compare the numerical solution with the analytical solution and empirical correlations.
- We will compare the velocity profile of the laminar flow and the turbulent flow.
- After finding the numerical solution, we will do some sampling and data manipulation.
- Then we will do some plotting (using gnuplot or Python) and scientific visualization.
- Remember, as we are introducing new closure equations for the turbulence problem, we need to define initial and boundary conditions for the new variables.
- We also need to define the discretization schemes and linear solvers to use to solve the new variables.
- It is also a good idea to setup a few functionObjects, such as: y+, minimum and maximum values, forces, time average, and online sampling.
- You will find the instructions of how to run this case in the file *`README.FIRST`* located in the case directory.

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- At this point, we are ready to run the simulation.
- To run the turbulent case, go to the **`GT1/turbulent`** directory.
- You can use the script `run_all.sh` to run the case automatically.
- Type in the terminal:
  - `$> sh run_all.sh`
- Feel free to open the file `run_all.sh` to see all the commands used.
- Or if you prefer, you can insert the commands manually in the terminal window (refer to the next slide).

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- Run the turbulent case by typing one command at a time.
- Go to the **`GT1/turbulent`** directory and type in the terminal:

```
1. $> foamCleanTutorials
2. $> blockMesh
3. $> extrudeMesh
4. $> createPatch -overwrite
5. $> foamRun | tee log
6. $> foamPostProcess -solver incompressibleFluid -func
   wallShearStress -noZero -noFunctionObjects -latestTime
7. $> foamPostProcess -solver incompressibleFluid -func yPlus
8. $> foamPostProcess -func sampleDict0 -latestTime
9. $> gnuplot gnuplot/gnuplot_script
10.$> paraFoam
```

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- In steps 1 we clean the case directory.
- Be careful, the command `foamCleanTutorials` will erase the mesh.
- If you want to erase the solution and keep the mesh, type in the terminal:
  - `$> foamListTimes -rm`
- In step 2, 3 and 4 we generate the mesh and rename and change the type of the patches.
- In step 5, we run the simulation using the modular solver command `foamRun`.
- In step 6, we use the utility postprocess to compute the wall shear stresses.
- In step 7, we use the utility postprocess to compute the y+ value at the walls.
- Steps 6 and 7 are very important when post processing turbulence simulations.
- In step 8 we do some sampling.
- In step 9, we plot the data sampled in step 8. For plotting, we use the program gnuplot.

Guided tutorials

From this point on and unless it is strictly necessary, or we introduce new features, we will not explain every single step or comment on the directories and files to be used.